

BIOS and POST Troubleshooting Theory

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

5 Common BIOS/POST Error Messages:-

BIOS/POST Error Message	Usually Indicates
1. CMOS Checksum Error	CMOS battery is weak/dead or BIOS settings are corrupted.
2. Keyboard Error / Keyboard Not Detected	Keyboard is disconnected, faulty, or USB/PS/2 port has a problem.
3. Memory Error / RAM Error	RAM is loose, faulty, or incompatible.
4. No Boot Device Found	HDD/SSD is not detected or boot configuration is incorrect.
5. CPU Fan Error	CPU cooling fan is disconnected, not working, or its speed is too low.

2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

AMI BIOS – 3 Beeps Error

Meaning: For classic AMI BIOS, 3 short beeps typically indicate a base-memory (RAM) read/write test failure the system cannot properly access the first 64 KB of memory.

Troubleshooting Steps:

1. **Reseat the RAM:** Turn off and unplug the PC, remove the RAM, clean/check the contacts, and firmly reinstall it.
2. **Test RAM individually:** If there are multiple RAM sticks, test them **one at a time** and, if necessary, try a different memory slot or a known-good compatible RAM module.

3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.

BIOS Version/Date	Insyde F.36, 03-02-2021
SMBIOS Version	3.2
Embedded Controller Version	89.38
BIOS Mode	UEFI
BaseBoard Manufacturer	HP
BaseBoard Product	86C1
BaseBoard Version	89.38

4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

Hint: Think about the CMOS battery and BIOS settings reset.

Meaning:

This error means the BIOS detected that the **CMOS settings/data are corrupted or have changed unexpectedly**. Therefore, the BIOS has loaded its **default settings** to allow the PC to start.

CMOS Checksum Error – Defaults Loaded:-

Step 1: Enter the **BIOS/UEFI setup** during startup (usually by pressing **Delete, F2, or F10**).

Step 2: Check the date, time, boot order, and other BIOS settings. Correct them if necessary.

Step 3: Save the settings and restart the PC.

Step 4: If the error appears again, switch off and unplug the PC and check/replace the **CMOS/RTC battery** (commonly a CR2032 coin-cell battery).

Step 5: After replacing the battery, enter BIOS again, set the correct date/time and required settings, then save and restart.